

Week 8: Options Market Making Theory and Dynamic Hedging

Avellaneda-Stoikov with a vector-valued inventory

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

Industry Mentor: Dendi Suhubdy (Bitwyre) · Guest: Fenni Kang (Coincall)

Learning Objectives

- ▶ Set up the El Aoud-Abergel multi-Greek framework
- ▶ Understand the inventory penalty as a quadratic form
- ▶ Quantify the P&L attribution of a hedged options book

Lecture 1: Multi-Greek Inventory & HJB

- ▶ Inventory becomes a vector of Greeks (delta, gamma, vega, ...)
- ▶ Quadratic inventory penalty and its economic meaning
- ▶ The multi-Greek HJB and separation ansatz
- ▶ Reduces to Avellaneda-Stoikov in the 1-D case

Lecture 2: Dynamic Hedging

- ▶ Continuous and discrete delta hedging
- ▶ The gamma-theta-variance identity (the key to options trading)
- ▶ External vs. integrated hedging architectures
- ▶ Perpetual futures as the hedge instrument; funding cost

Recitation & Problem Set 8 (Python/PyTorch)

- ▶ Multi-Greek HJB walk-through; PyTorch hedged-book simulation
- ▶ PS8: inventory dynamics, penalty calibration, threshold hedger
- ▶ Reading: El Aoud-Abergel (2015); Taleb (1997)